

NUMERICAL INVESTIGATION OF SHOCK-INDUCED DEFORMATION OF BUBBLE-LADEN DROPLETS

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Presentation Outline

- **Motivation & Research Question**
- **Background & Knowledge Gaps**
- **Methodology**
- **Clean Baseline Results**
- **Gas-Seeded Parametric Results**
- **Conclusions & Future Work**

Motivation – Rain Erosion

- Hypersonic glide vehicles encounter atmospheric hydrometeors during terminal descent at 0–7 km
 - Micron cloud droplets to millimeter raindrops
- Cumulative droplet impacts roughen the surface → early boundary-layer transition → heating/drag/control-authority penalties
- These margins are already thin in air-breathing systems

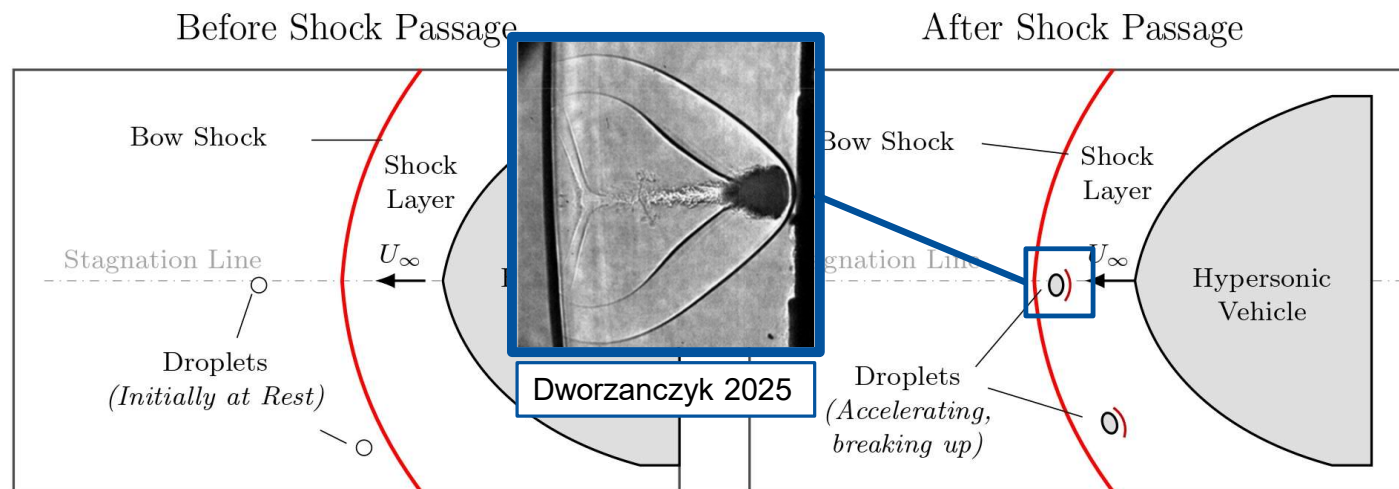


Figure: Schematic of atmospheric droplets interacting with the bow shock of a hypersonic vehicle before and after shock passage. Adapted from Aguilera 2023.

Motivation – Rain Erosion

- Impact loading scales with $\sim \rho D^2 V^2$
- A single 2mm raindrop can exert up to 40kN of force

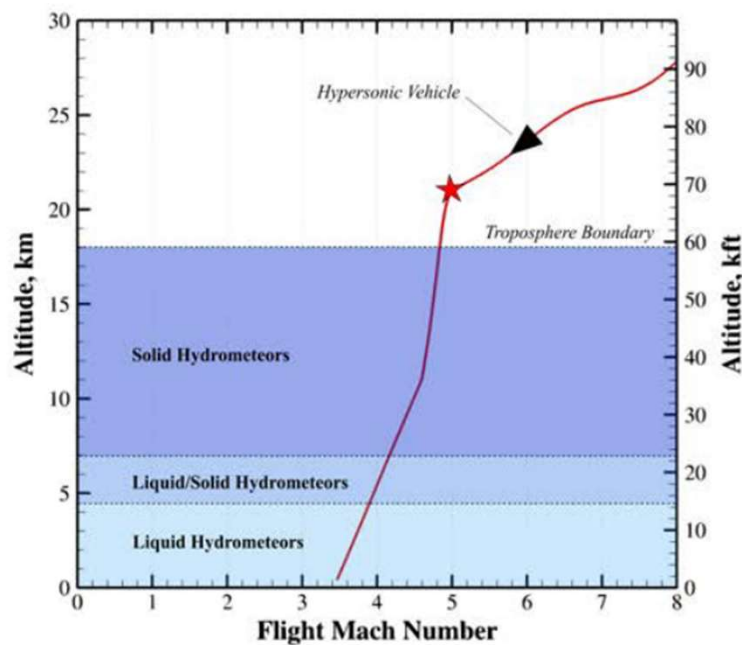


Figure: View of hypersonic vehicle terminal dive phase (Aguilera 2023).



Figure: 4mm droplet impact on radome at Mach 3 (Moylan 2013).

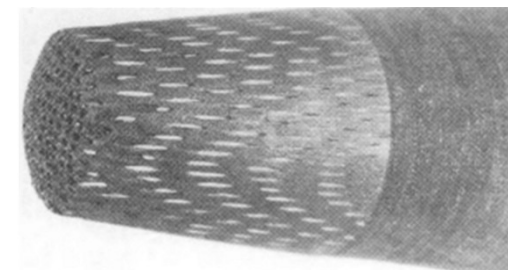


Figure: Vehicle nosetip post flight through clouds (Sandia National Labs).

Motivation – Drop Acoustics and Rain Formation

- Shock transmission → internal wave dynamics
 - Reflections / expansion waves
- Cloud droplets and raindrops form through condensation on suspended aerosol particles
 - May trap microscopic gas pockets (heterogenous nuclei)

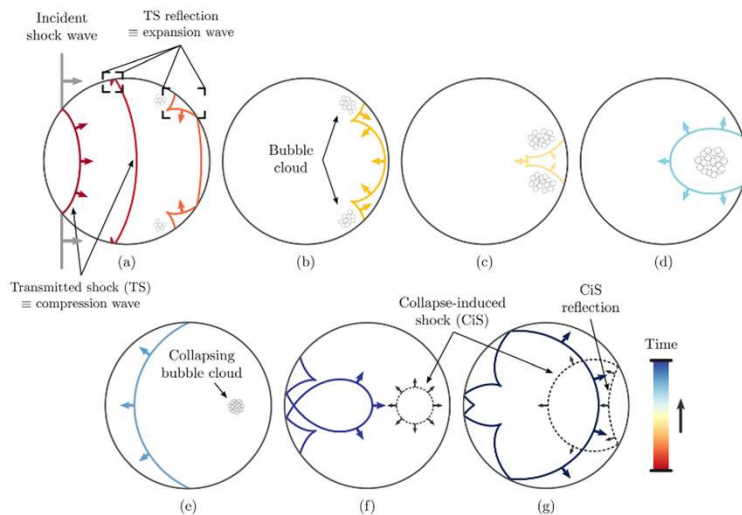


Figure: Progression of droplet internal waves (Biasiori-Poulanges 2023).

Particle	Diameter, μm	Terminal velocity, m s^{-1}
Typical condensation nucleus	0.2	1.0×10^{-6}
Typical cloud drop	20	0.01
Cloud drop / raindrop conventional borderline	100	0.27
Typical raindrop	2000	6.50

Table: Typical sizes and terminal velocities of liquid particles in clouds (Aguilera 2023).

Scope and Objectives

Can the consideration of pre-existing nuclei measurably affect droplet deformation?

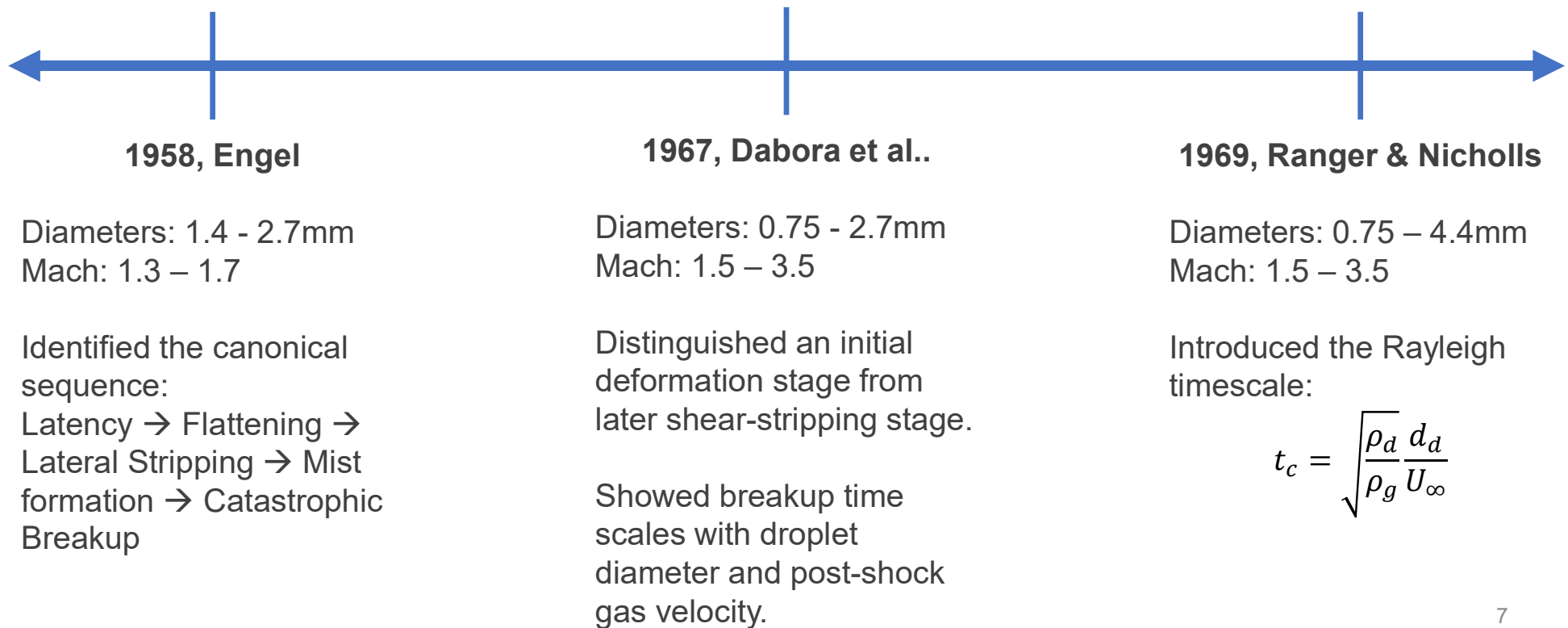
Computationally study the role of pre-existing **non-condensable gas nuclei** in the **early-time** response of a shock-loaded liquid droplet.

Objectives:

1. Develop and verify a cylindrical Euler-Lagrange bubble model
2. Benchmark the compressible multiphase framework
3. Characterize the “clean” shock-droplet baseline
4. Quantify the mechanical response of embedded gas nuclei
5. Assess the impact of the bubble response on the droplet pressure field and deformation

Literature: Classical Shock-Droplet Breakup

Early shock-tube studies established the canonical macroscopic breakup sequence and the characteristic breakup timescale.



Literature: Classical Shock-Droplet Breakup

Early shock-tube studies established the canonical macroscopic breakup sequence and the characteristic breakup timescale.

1969, Reinecke & McKay

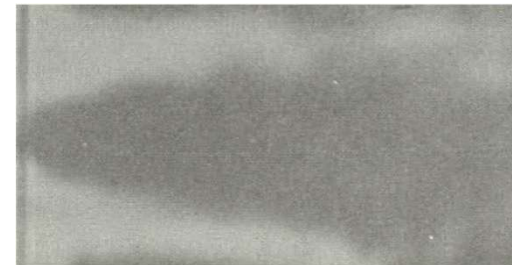
Diameters: 0.5 – 2.5mm

Mach: 3, 6, 9, 11, 12

Discovered breakup mode transition at ~Mach 6.

< Mach 6: Stripped by aerodynamic shear (KH)

> Mach 6: Catastrophic shattering (RT piercing)



*Figure: Droplet, Mach 3
(Reinecke-McKay 1969)*



*Figure: Droplet, Mach 11
(Reinecke-McKay 1969)*

Literature: Cavitation and Entrained Nuclei

Internal pressure focusing can activate cavitation or gas nuclei, but existing studies have not isolated the mechanical response of pre-existing non-condensable nuclei.

Sembian et al., 2016

- Provided direct shock-water-column evidence of internal wave focusing and cavitation

Forehand et al., 2023-2024

- Compared simulations with and without cavitation modeling
- One-way coupling bubble dynamics

Shannon 2024

- Resolved one cavitation bubble near an interface and showed it can seed KH roll-up

Briggs / Aguilera, 2023-2026

- Experiments of degassed, ambient, and doped droplets ($\sim$ Mach 3, $D=1-3\text{mm}$)
- Limited diagnostic, little evidence

Cavainolo 2026

- One-way cavitation estimate from CFD pressure history

Literature: Knowledge Gaps

Prior work identified cavitation-prone regions and droplet breakup mechanisms, but did not resolve the direct feedback of pre-existing gas nuclei.

Gap 1:

Isolated nuclei response

Gap 2:

Two-way coupled multi-nuclei effects

Gap 3:

Early pressure field feedback

Hypothesis: mechanically responding NCG nuclei modify early internal pressure dynamics before producing visible changes in droplet deformation.

Methodology: Computational Setup

All simulations are performed in cylindrical coordinates (x, r) under an axisymmetric assumption.

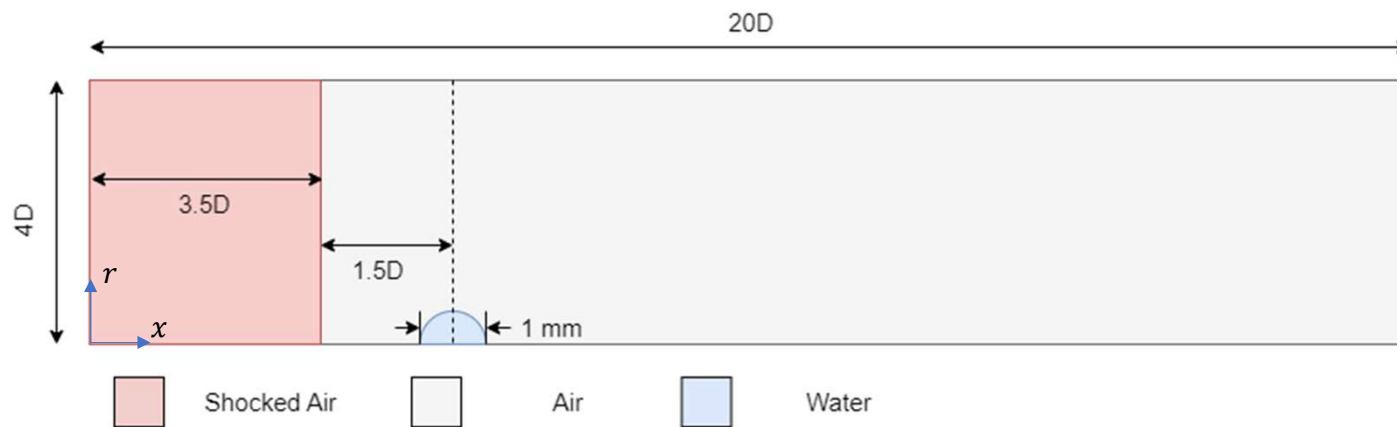


Figure: Computational domain, normalized by droplet diameter, D .

Boundary conditions:

- The bottom boundary is an axis of symmetry
- All other boundaries are non-reflecting boundaries

Initial conditions:

- Shocked air set by Rankine-Hugoniot jump conditions into quiescent, ambient air

Methodology: Simulation Matrix

Parameter space: For a 1mm droplet, $M_s \in \{3, 5, 7\}$, $R_0 \in \{0.1, 0.2, 1, 2\} \mu m$, $N_b \in \{10, 100, 1000\}$

- A clean case is run at each Mach, 36 doped cases

Parameter	Values			
Shock Mach, M_s	3	5	7	-
Initial bubble radius, R_0 (μm)	0.1	0.2	1	2
Number of bubbles, N_b	10	100	1000	-

Table 3.3 Shock-droplet simulation matrix. This spans the doped cases.

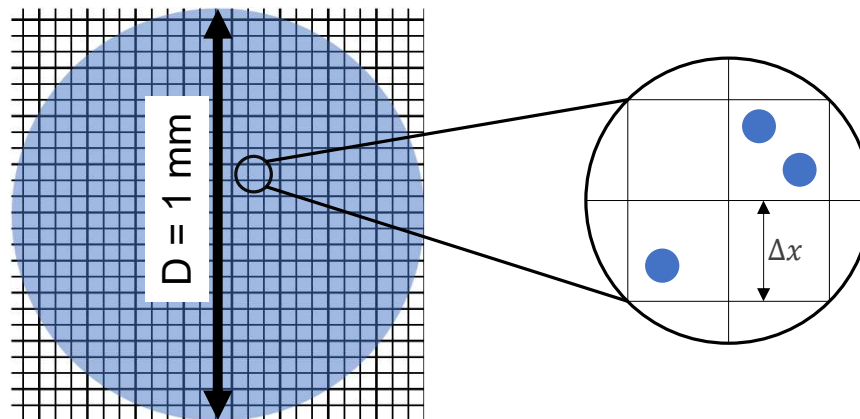


Figure: Sub-grid Lagrangian bubble representation.

● : Sub-grid Lagrangian bubble

Methodology: Number Density and Void Fraction

Number density:

$$n = \frac{N_b}{V_{drop}}, \quad V_{drop} = \frac{4}{3}\pi \left(\frac{D}{2}\right)^3$$

- For $N_b = 10, 100, 1000$, n yields $19, 191, 1910 \text{ mm}^{-3}$, respectively

Void Fraction:

$$\beta_0 = \frac{N_b \frac{4}{3}\pi R_0^3}{V_{drop}}$$

	R_0			
N_b	$0.1 \mu m$	$0.2 \mu m$	$1 \mu m$	$2 \mu m$
10	8×10^{-11}	6.4×10^{-10}	8×10^{-8}	6.4×10^{-7}
100	8×10^{-10}	6.4×10^{-9}	8×10^{-7}	6.4×10^{-6}
1000	8×10^{-9}	6.4×10^{-8}	8×10^{-6}	6.4×10^{-5}

Table 3.4 Initial global gas void fraction β_0 for the bubble-loading parameter sweep.

Methodology: The Numerical Model

The present simulations are performed in the Multi-Component Flow Code (MFC)

- Open-source, high-order, compressible multiphase solver
- Sub-grid bubble models
- Highly parallel

Resolved Eulerian phase interface:

- Reduced five-equation diffuse-interface formulation of Allaire et al. 2002.

$$\frac{\partial(\alpha_i \rho_i)}{\partial t} + \nabla \cdot (\alpha_i \rho_i \mathbf{u}) = 0, \quad i = 1, 2, \dots, N$$

$$\frac{\partial(\rho \mathbf{u})}{\partial t} + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u} + p \mathbf{I}) = 0$$

$$\frac{\partial(\rho E)}{\partial t} + \nabla \cdot [(\rho E + p) \mathbf{u}] = 0$$

$$\frac{\partial \alpha_i}{\partial t} + \mathbf{u} \cdot \nabla \alpha_i = K \nabla \cdot \mathbf{u}$$

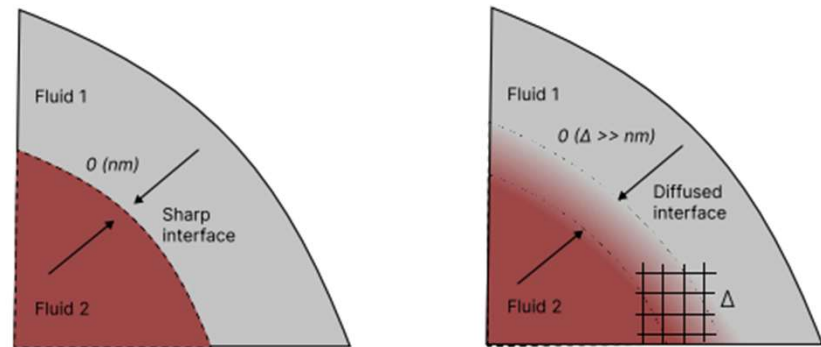


Figure: Sharp and diffuse interface of a multiphase flow (Adebayo 2025).

Methodology: The Numerical Model

Each material is closed with a stiffened-gas equation of state

$$p_i = (\gamma_i - 1)\rho_i e_i - \gamma_i \pi_{\infty,i}$$

The material properties are listed below:

Quantity	Air / bubble gas	Liquid water
Density, ρ	1.17 kg m ⁻³	1000 kg m ⁻³
Ratio of specific heats, γ	1.4	4.4
Stiffened pressure, π_{∞}	0	600 MPa

Table: Equation-of-state and material parameters used in the shock-droplet simulations.

Viscous stresses and surface tension are neglected in the resolved Eulerian equations.

Methodology: Euler-Lagrange Bubble Model

The doped-droplet cases contain N_b Lagrangian gas nuclei embedded inside the initially liquid droplet.

- Each nucleus is modeled as a spherical sub-grid bubble
- The bubbles do not advect
- The resolved Eulerian field provides the local driving pressure
- Bubble volume change feeds back to the Eulerian field through source terms

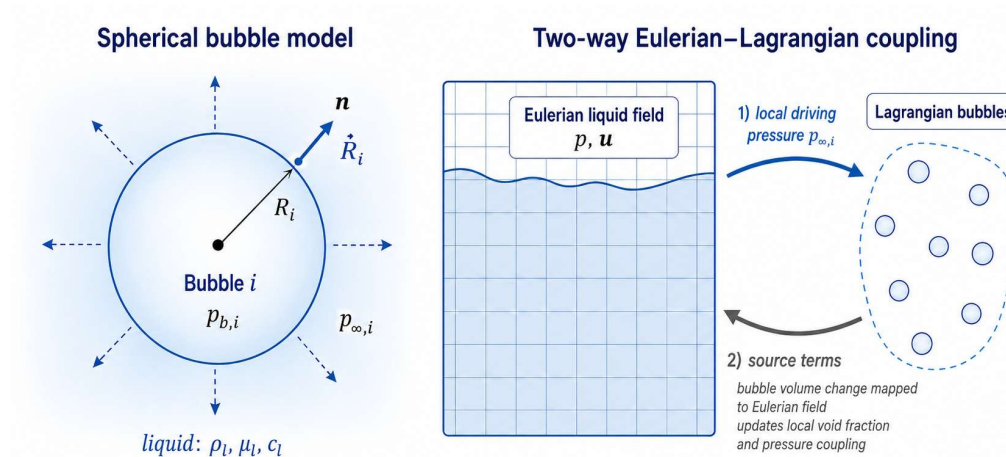


Figure: Sub-grid spherical bubble model and two-way Eulerian-Lagrangian coupling used to evolve the embedded gas nuclei.

Methodology: Keller-Miksis-Type Model

The sub-grid bubble dynamics:

- Bubble radius, $R_i(t)$, is evolved with a Keller-Miksis-type equation

$$\begin{aligned}
 & \underbrace{\left[\left(1 - \frac{\dot{R}_i}{c_l} \right) R_i + \frac{4\mu_l}{\rho_l c_l} \right] \ddot{R}_i}_{\text{Liquid inertia + compressibility/viscous correction}} + \underbrace{\frac{3}{2} \left(1 - \frac{\dot{R}_i}{3c_l} \right) \dot{R}_i^2}_{\text{Nonlinear radial inertia}} \\
 &= \underbrace{\frac{1}{\rho_l} \left(1 + \frac{\dot{R}_i}{c_l} \right) \left(p_{b,i} - p_{\infty,i} - \frac{4\mu_l \dot{R}_i}{R_i} - \frac{2\sigma}{R_i} \right)}_{\text{Net pressure driving, corrected for finite sound speed}} + \underbrace{\frac{R_i}{\rho_l c_l} \left(\dot{p}_{b,i} + \frac{4\mu_l \dot{R}_i^2}{R_i^2} + \frac{2\sigma \dot{R}_i}{R_i^2} \right)}_{\text{Time-varying pressure/ acoustic radiation correction}}
 \end{aligned}$$

- Extends the Rayleigh–Plesset equation (RPE) by retaining first-order liquid compressibility effects
 - As $c_l \rightarrow \infty$, the equation reduced to RPE
- Appropriate for spherical sub-grid nuclei in a weakly compressible liquid

Methodology: Numerical Methods

The Eulerian equations are discretized on a structured grid using a finite-volume formulation.

Numerical Component	Method Used
Face reconstruction	2 nd order MUSCL + van Leer limiter
Interface treatment	THINC interface sharpening
Riemann Solver	HLLC
Time-integration	SSP-RK3 (TVD)
Flow-bubble coupling	Operator splitting

Table 3.2 Summary of numerical methods used for the Eulerian flow solver and sub-grid bubble dynamics.

A constant CFL value of 0.1 is used in all simulations.



Figure: Numerical results for a planar Mach onto a gas bubble. With THINC (left) and without THINC (right) are both shown (Shyue-Xiao 2014).

Methodology: Code Extensions

Cylindrical Euler-Lagrange Bubble Model: allows Lagrangian gas nuclei to be used in the axisymmetric formulation used for the simulations

- Azimuthal mapping operator to each 3D bubble:

$$T_{\theta}(\cdot) = \frac{1}{2\pi r} \int_0^{2\pi} (\cdot) r d\theta$$

- Preserves equivalent axisymmetric bubble volume
- Deposits bubble source terms consistently in cylindrical coordinates
- Resulting axisymmetric void fraction field:

$$\beta_{axi}(z, r) \equiv (T_{\theta}\beta)(z, r) = \sum_{n=1}^{N_b} V_n \delta_{axi}(z, r; z_n, r_n)$$

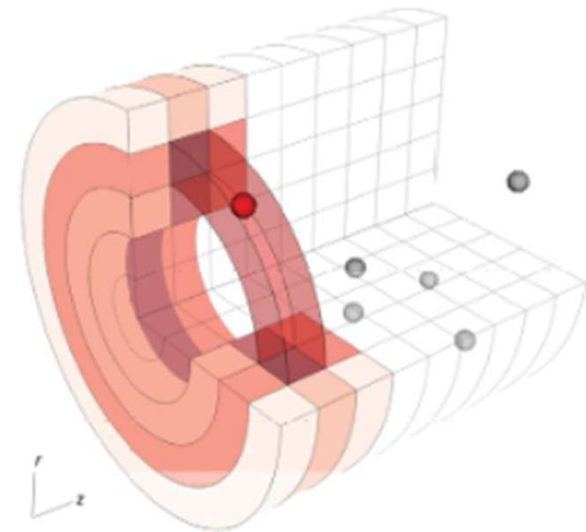


Figure: Axisymmetric mapping of a 3D Lagrangian bubble kernel onto neighboring FVM cells (Maeda-Colonius 2018).

Methodology: Code Extensions

Sub-Grid Stochastic Closure: in a reduced simulation, the Eulerian field carries only the dimension-averaged pressure, but the true pressure fluctuates.

- True local pressure:

$$p(x_n) = \bar{p} + p'_n$$

Problem: MFC identified simulations as “2D” when there are no z-direction grid points

- This condition is true for both 2D Cartesian and axisymmetric
- As a result, the 2D Cartesian stochastic correction was also applied to axisymmetric cases

Fix implemented:

- Added a cylindrical-coordinate branch check:
 - Apply correction only if 2D and not cyl_coord
- Ensures the axisymmetric bubble simulations use the intended source-mapping without an inconsistent Cartesian stochastic correction

Methodology: Code Extensions

Cylindrical Bubble Verification: verified the implementation with a bubble-screen configuration

- Single-cycle sinusoidal acoustic pulse ($f = 300 \text{ kHz}$, amplitude $p_a = 10^5 \text{ Pa}$)
- Propagates through water onto a population of $N = 200$ bubbles of $R_0 = 10 \mu\text{m}$
 - Initial void fraction $\beta_0 \approx 9.3 \times 10^{-6}$ is randomly distributed
- Run in full 3D and axisymmetric

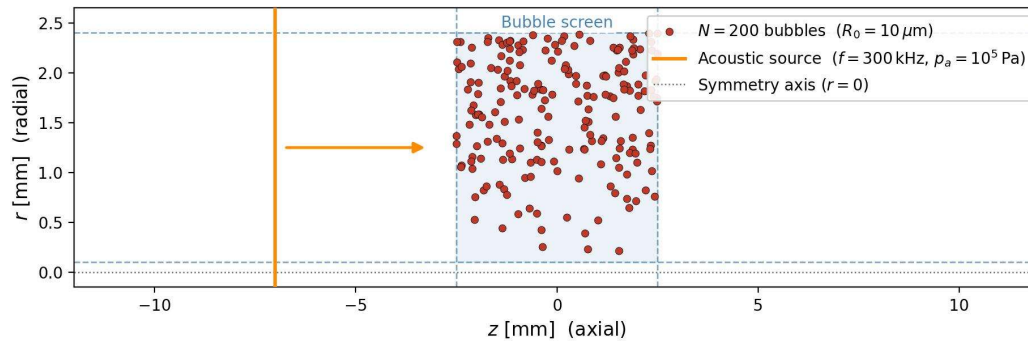


Figure: Computational domain for the cylindrical bubble screen test. Acoustic source upstream pulses for a single cycle toward a bubble screen.

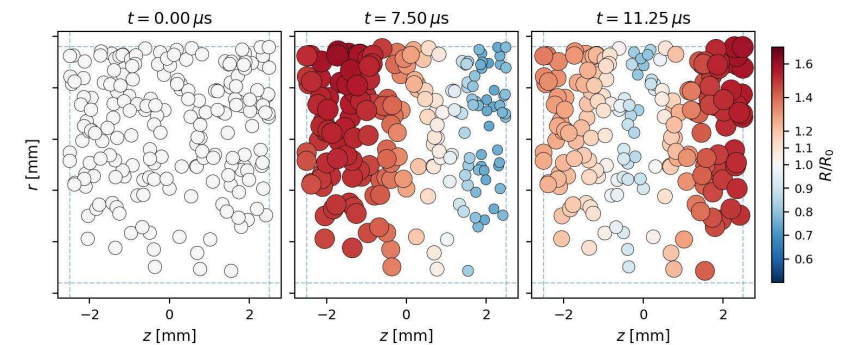


Figure: Bubble screen at three time instants. Circle areas scale with $(R/R_0)^2$. Red indicates and expanded bubble, white indicates equilibrium, and blue indicates compressed.

Methodology: Code Extensions

Cylindrical Bubble Verification: verified the implementation with a bubble-screen configuration

- Compared between the full 3D and the axisymmetric implementations
- Ensemble averaged over 10 realizations
- Comparisons show good agreement over the duration of the first pulse

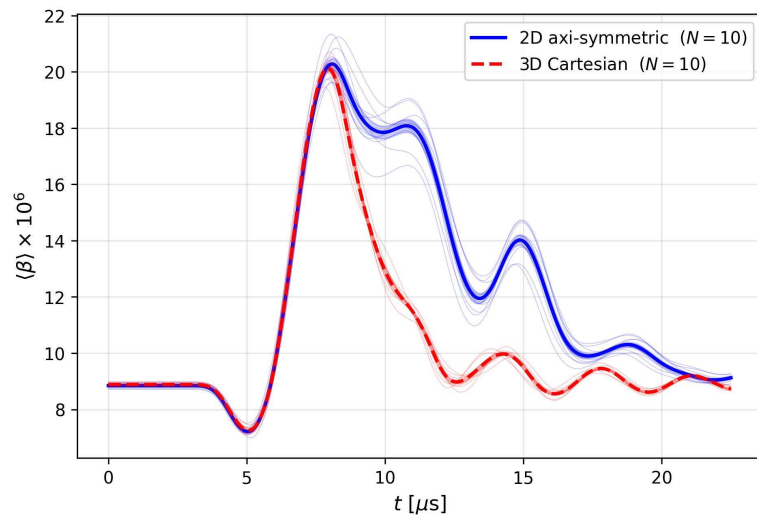


Figure: Ensemble averaged void fraction of the 2D axisymmetric and full 3D Cartesian bubble screen tests.

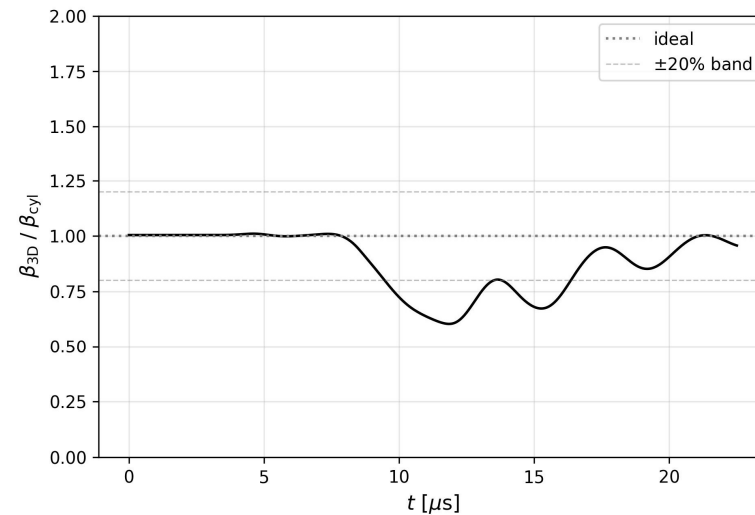


Figure: Ratio of ensemble averaged void fractions.

Methodology: Code Extensions

Multi-Fluid THINC Extension: implementation in MFC originally contained a two-fluid assumption

- Old method assumed that after THINC interface reconstruction, $\alpha_2 = 1 - \alpha_1$
- New version:
 - Computes the ratio of the new “remaining volume” to the old one

$$S_{other} = \frac{1 - \alpha_{THINC}}{1 - \alpha_{THINC,old}}$$

- Rescales all non-tracked fluids proportionally (conserving sum of volume fractions)
- Recall: THINC is sharpening **one interface** (water-air)
- Since nothing physically mixed / separated air from bubble-gas, the relative proportions need to be maintained
- A different approach would need to be taken if the bubbles were resolved

Methodology: Benchmarking

Benchmarking Against Shock-Water Column Experiments:

- Sembian 2016 studies the interaction of a planar shock (Mach = 1.75 and 2.4) onto a droplet column of diameter $D_0 = 22 \text{ mm}$
- Large droplet allows for the observation of key internal acoustics
- Numerical schlieren is compared directly to the experimental shadowgraph
 - Comparison of transmitted shocks, reflected waves, expansion regions, slip surfaces, and triple-point/Mach-stem structures
- Excellent agreement is observed

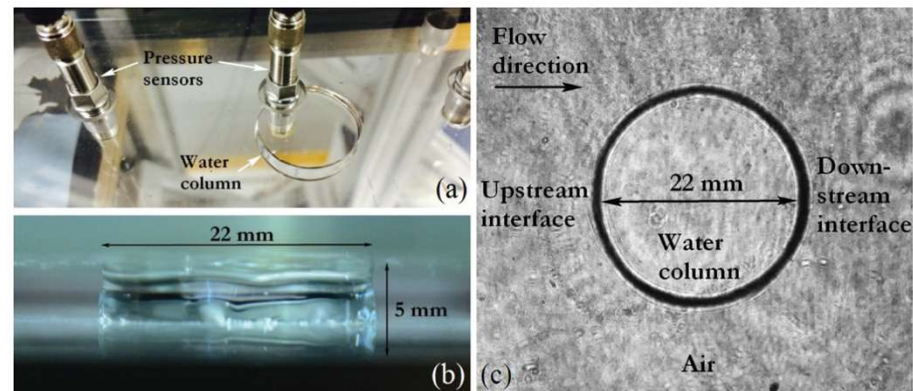


Figure: 22mm water column of Sembian 2016.

Methodology: Validation / Benchmarking

Benchmarking Against Shock-Water Column Experiments:

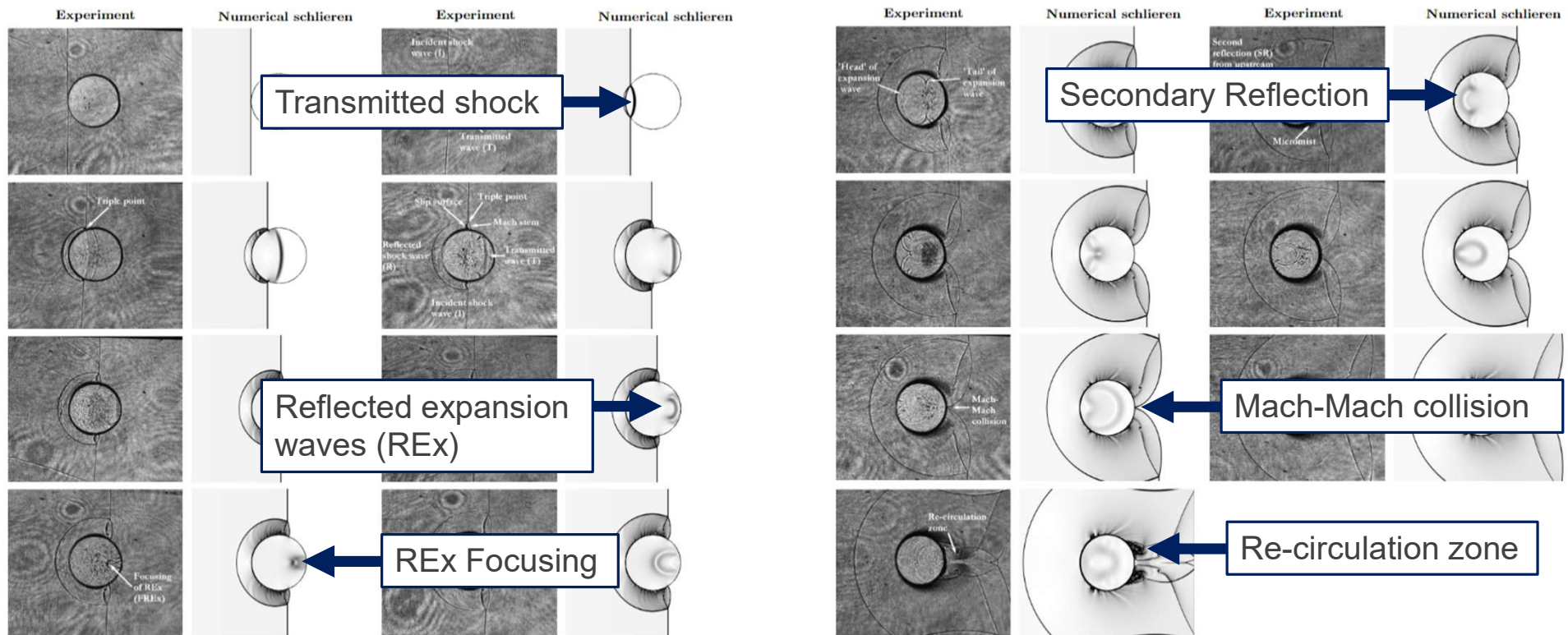


Figure: Comparison between experimental shadowgraph images from the Mach 2.4 shock-water-column experiment of Sembian et al. and numerical schlieren fields from the MFC simulation.

Methodology: Mesh Independence

- Performed on a clean droplet in our case of interest
- Mach 7 is chosen for the study as it produces the largest pressure gradients within the droplet
- A refinement factor of $\sim\sqrt{2}$ is used

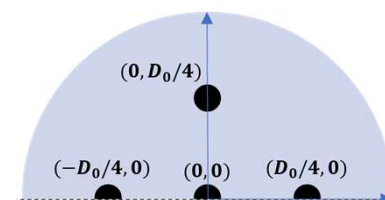
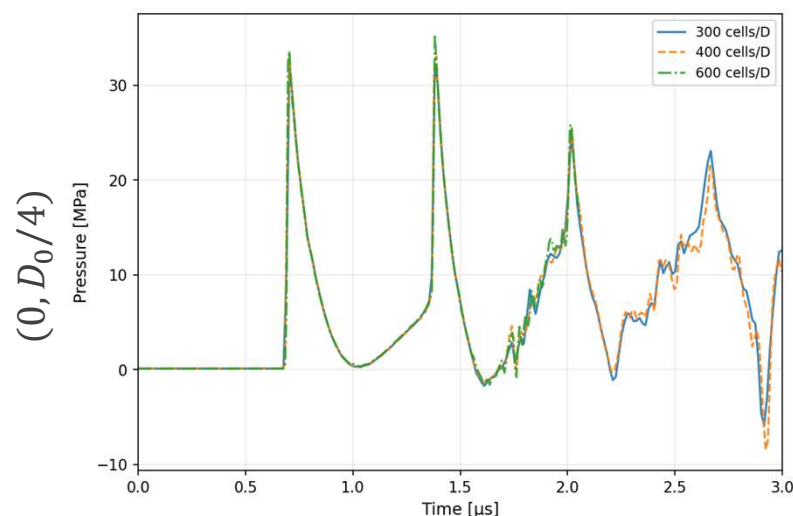
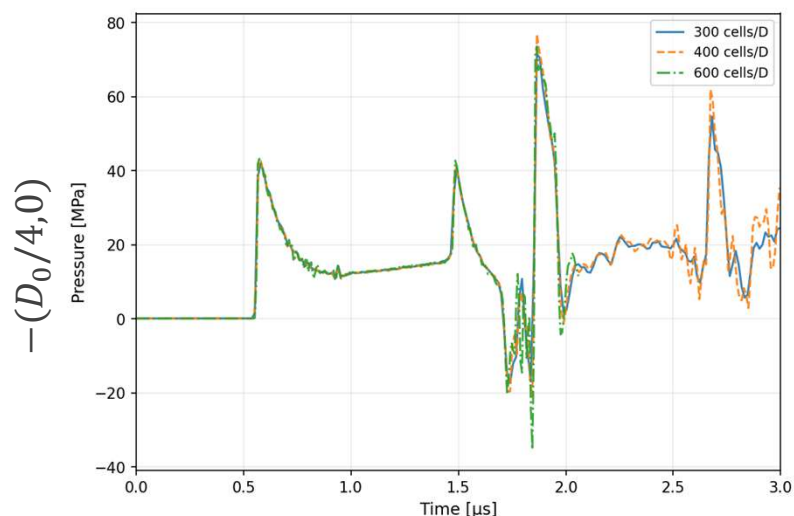
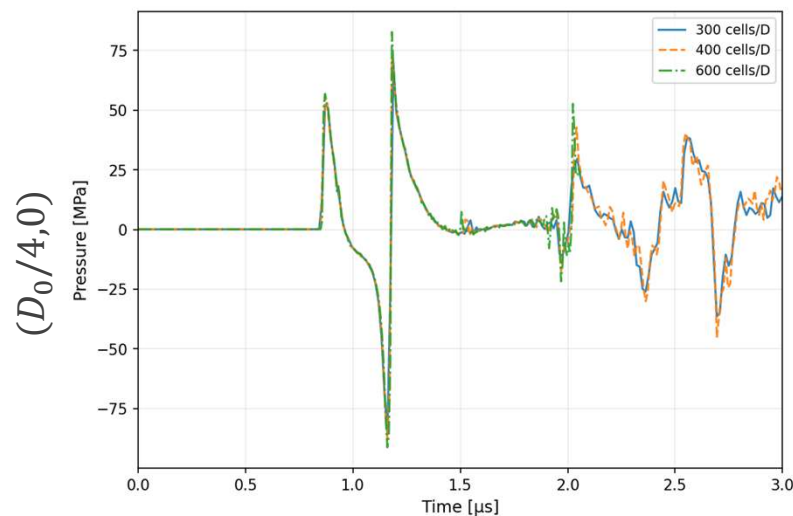
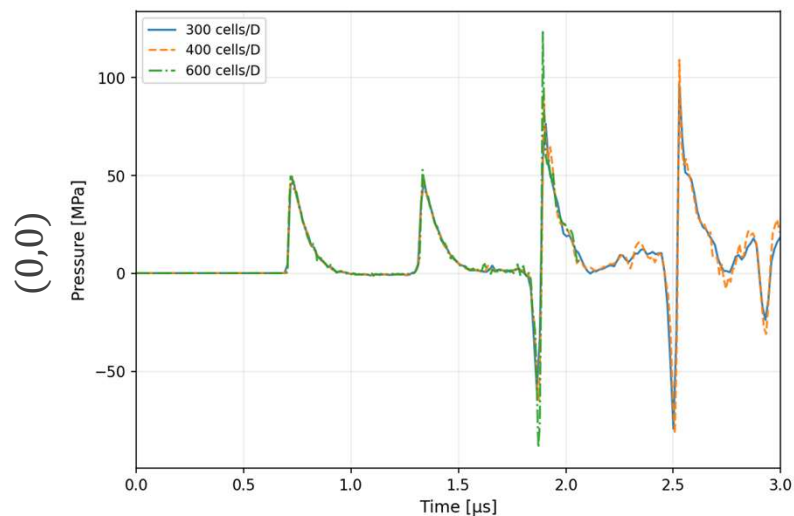
Mesh	Cells / D_0	Grid size ($N_x \times N_r$)	Δx
Coarse	300	6000×1200	$3.33 \mu m$
Medium	400	8000×1600	$2.50 \mu m$
Fine	600	12000×2400	$1.67 \mu m$

Table 3.5 Mesh levels used in the shock-droplet mesh-independence study.

- Shock-arrival time, peak pressure, pressure decay, and area-averaged pressure response across mesh levels was compared
- All meshes show close agreement, hence the medium mesh was chosen

Methodology: Mesh Independence

Nearest-cell pressures

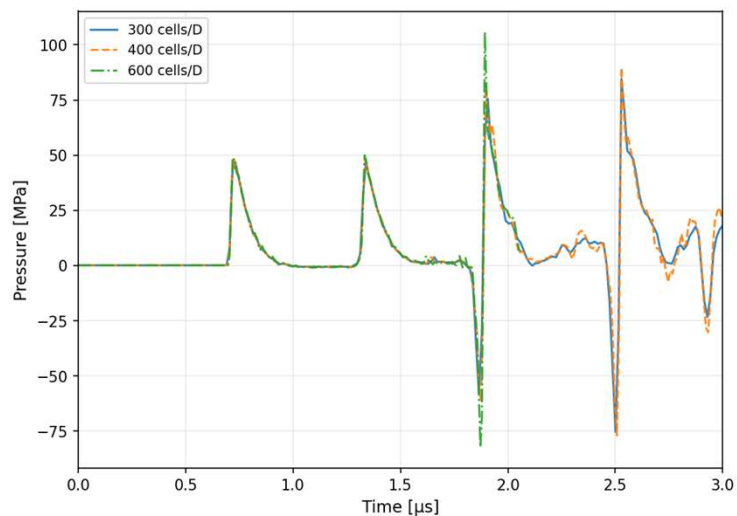


Methodology: Mesh Independence

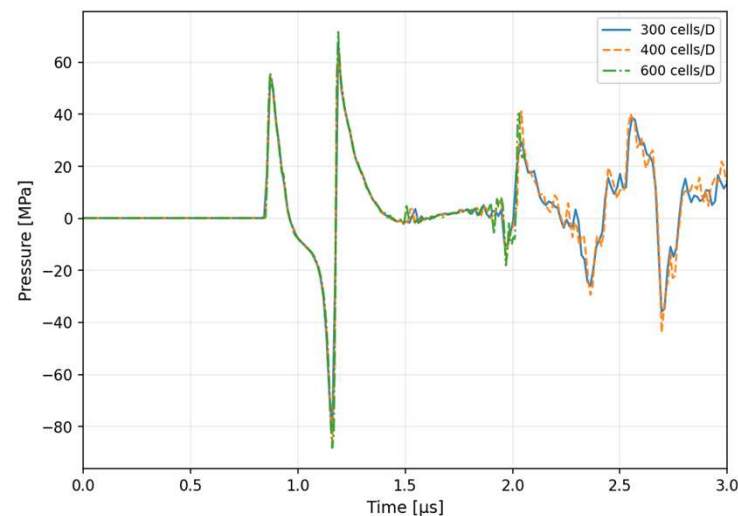


Area-averaged pressures ($D_{avg} = D_0/40$)

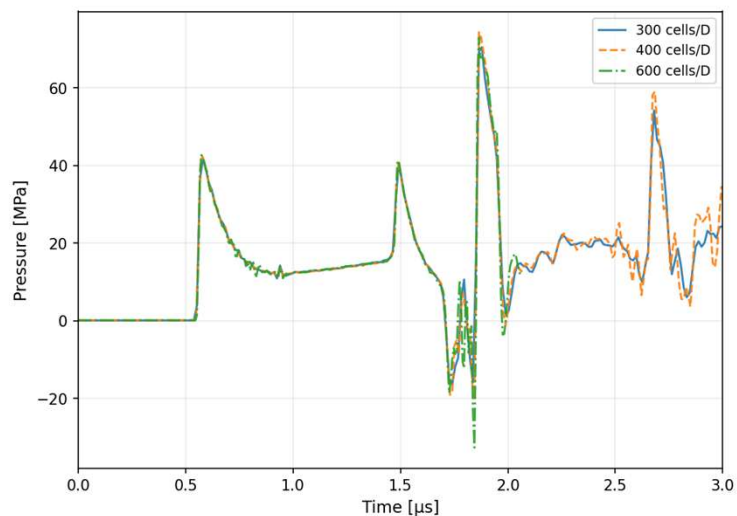
(0,0)



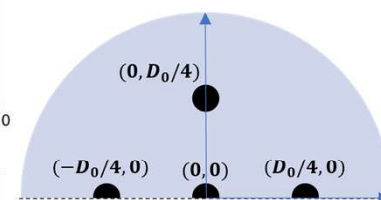
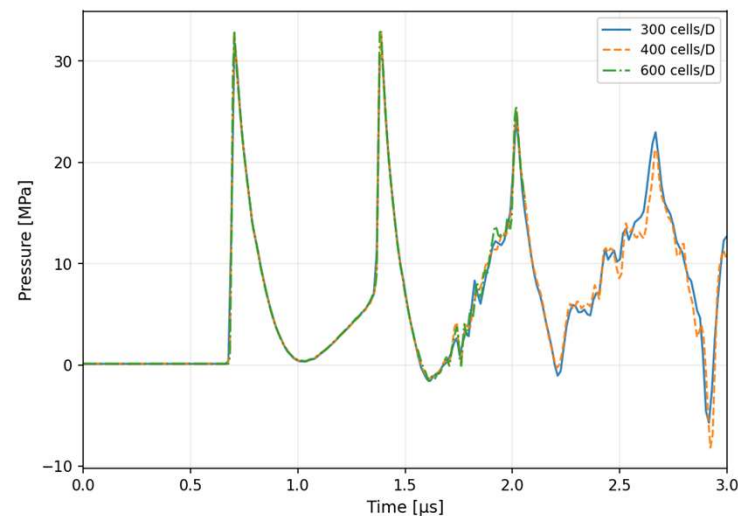
($D_0/4, 0$)



$-(D_0/4, 0)$



(0, $D_0/4$)



Methodology: Summary of Assumptions

Geometry / interface:

- Axisymmetric geometry
- Diffuse-interface formulation with interface sharpening
- Droplet-scale surface tension neglected

Sub-grid nuclei model:

- Bubbles are modeled as spherical sub-grid nuclei
- Bubbles remain fixed in space; no stochastic correction
- No asymmetrical collapse
- Bubble-bubble interaction only through the resolved pressure field

Thermodynamic / transport simplifications

- Non-condensable gas nuclei
- No vapor production, mass transfer, and phase change
- Polytropic bubble gas response
- Neglecting viscosity in Eulerian phase

Clean Results: Non-dimensionalization

Shock-droplet interaction in the absence of dissolved gas nuclei.

- All quantities are presented in the dimensionless time:

$$\tau = \frac{t - t_{arr}}{t_b}$$

- Where t_b is the Ranger-Nichols aerobreakup timescale:

$$t_b = \frac{D}{U} \sqrt{\frac{\rho_l}{\rho_g}}$$

- Define acoustic crossing time:

$$t_c = D/c_w$$

$$\tau_c = t_c/t_b$$

Quantity	$M_s = 3$	$M_s = 5$	$M_s = 7$
$p_{post} [MPa]$	1.044	2.929	5.757
$u_{post} [m s^{-1}]$	772.5	1390.6	1986.5
$t_b [\mu s]$	19.27	9.40	6.31
$t_c = D/c_w [\mu s]$	0.615	0.614	0.613
$\tau_c = t_c/t_b$	0.0319	0.0653	0.0971

Table: Key acoustic timescales.

Clean Results: Internal Wave Structure

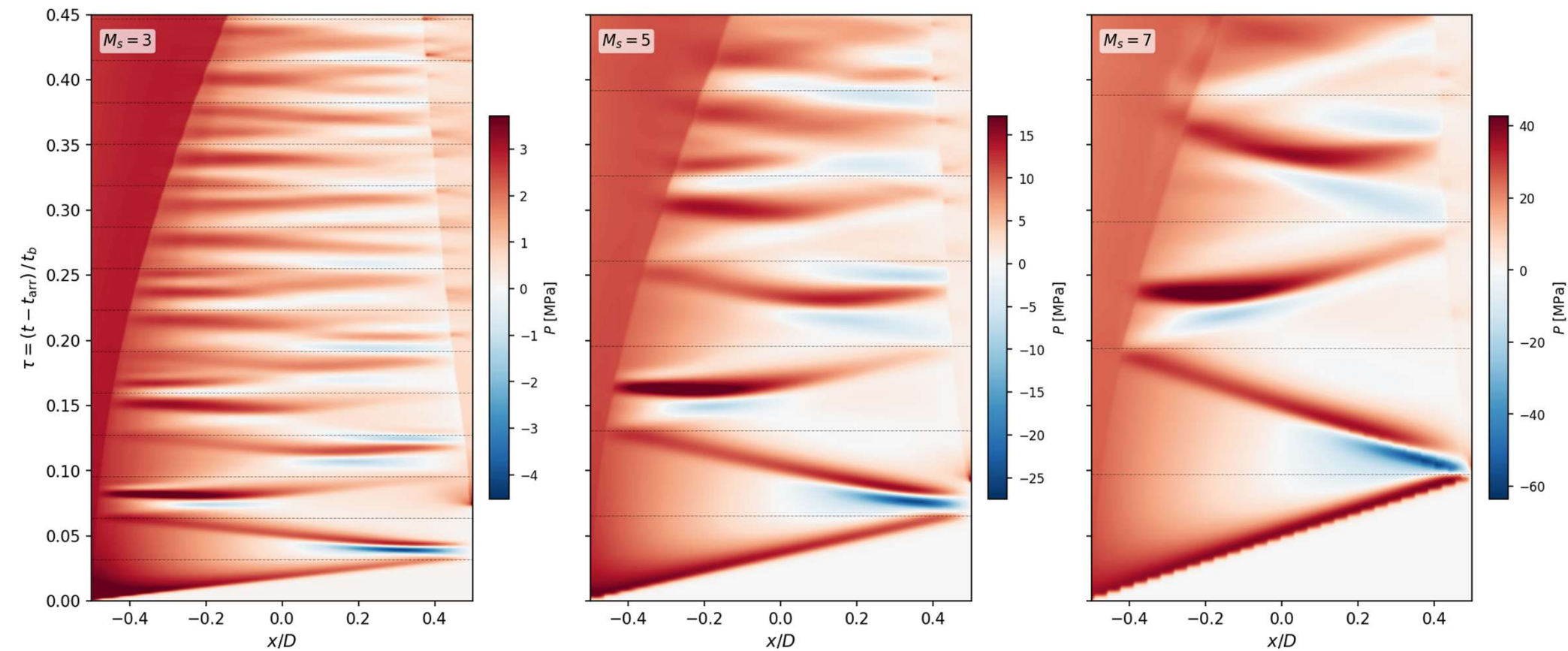


Figure: Space-time diagram of axial centerline pressure for the three clean-baseline cases. Left: Mach 3; center: Mach 5; right: Mach 7.

Clean Results: Peak Pressure

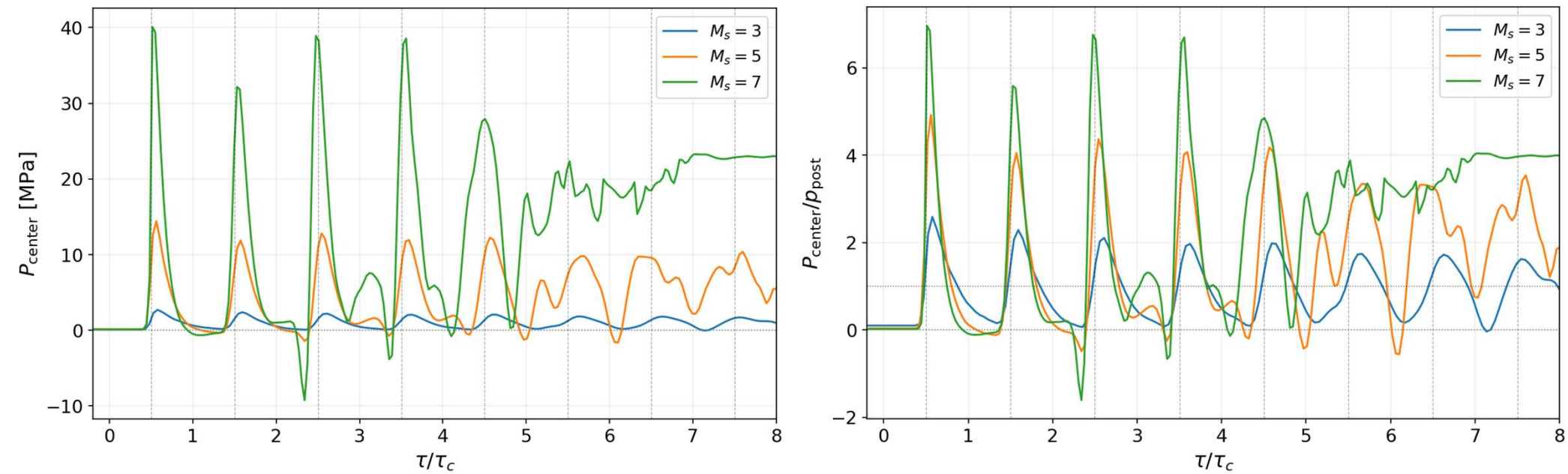


Figure: Center-probe pressure history plotted against time normalized by the acoustic crossing time. *Left:* absolute pressure. *Right:* pressure normalized by case-specific post-shock pressure.

Clean Results: Rarefaction

Each acoustic reflection generates a rarefaction wave that propagates back into the droplet and produces tensile pressures.

Quantity	$M_s = 3$	$M_s = 5$	$M_s = 7$
$P_{cl,min} [MPa]$	-4.53	-27.5	-63.7
$P_{cl,min}/p_{post}$	-4.3	-9.4	-11.1
x/D at $P_{cl,min}$	0.33	0.34	0.38
τ at $P_{cl,min}$	0.038	0.076	0.106
τ_{min}/τ_c	1.19	1.17	1.09

Table: Minimum centerline pressure. $P_{cl,min}$ is the global minimum over the centerline.

- Non-dimensional acoustic crossing times reveal the minimum rarefaction on the centerline occur shortly after the first expansion wave reflection

Clean Results: Rarefaction

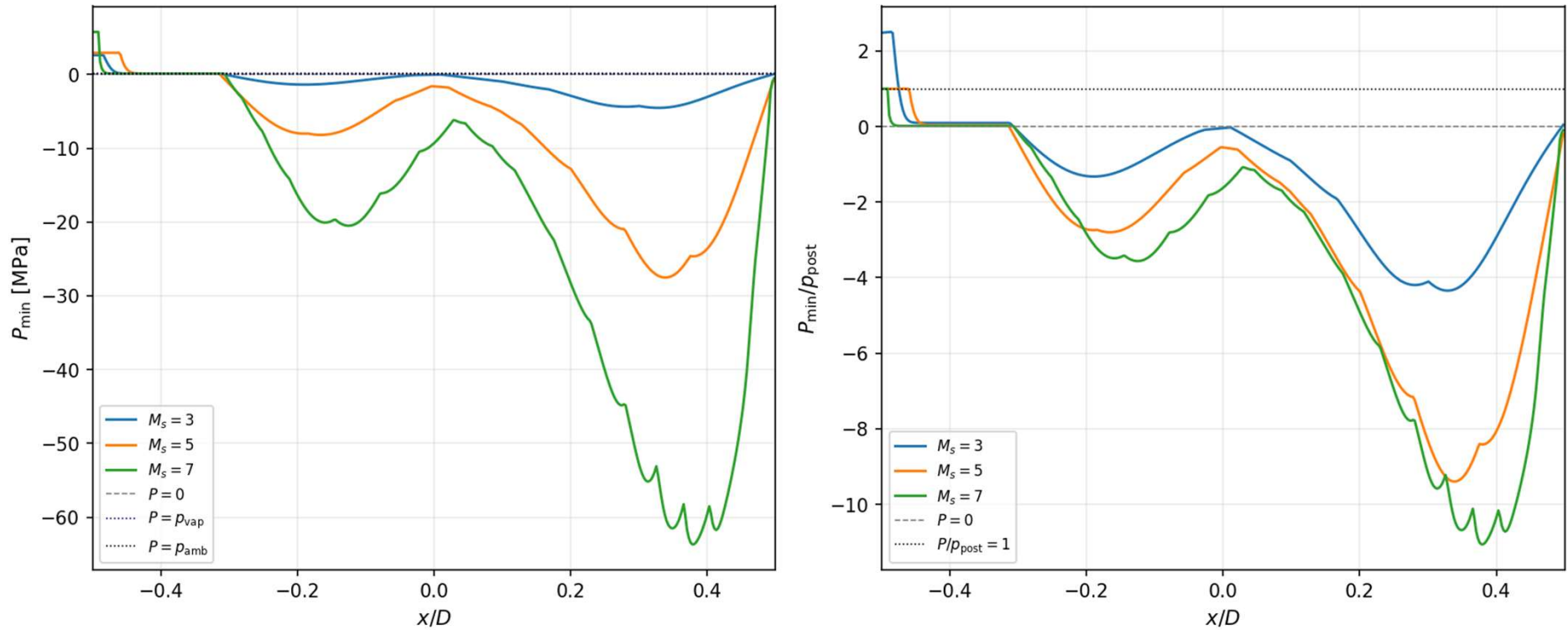


Figure: Minimum centerline pressure along the droplet axis, minimized over $\tau \in [0, 0.45]$. *Left*: absolute minimum pressure. *Right*: normalized by p_{post} .

Clean Results: Underpressure Integral

To quantify the tensile loading experienced by a nucleus, we define the normalized underpressure integral:

$$\hat{I}(x; p_{cav}) = \frac{1}{p_{post} t_b} \int_0^{\tau_{end}} \max(0, p_{cav} - P(x, \tau)) d\tau$$

- Concentrated in downstream half
- Increases with Mach at every position

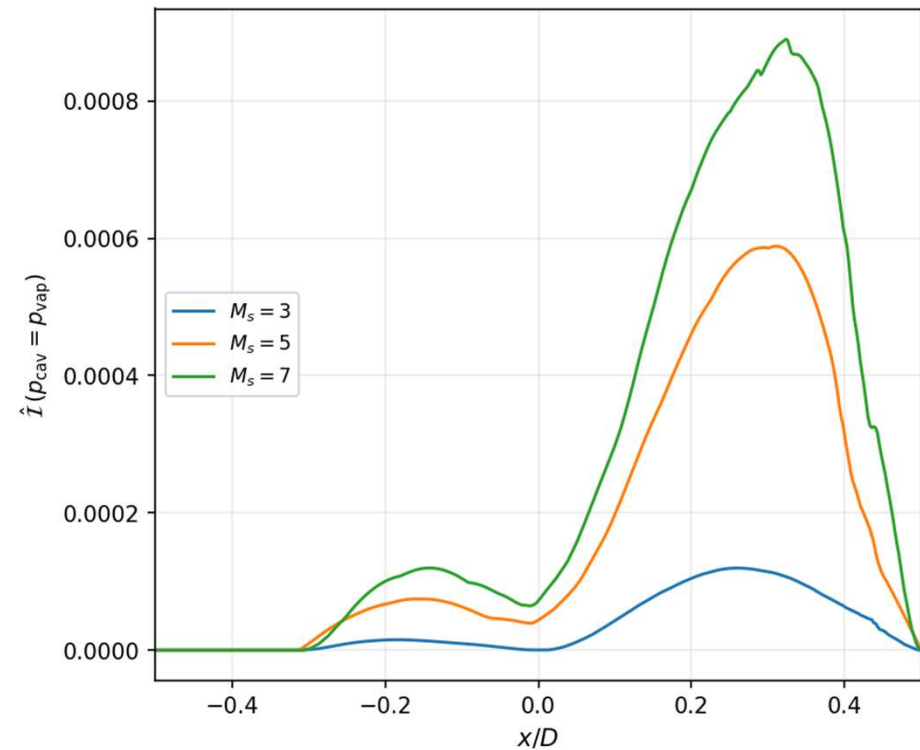


Figure: Underpressure integral at centerline over Mach = 3, 5, 7.

Clean Results: Early Droplet Deformation

Early-time droplet deformation:

- Noticeable flattening on the windward face
- Rim roll-up and lip formation
- RTP mode formation
- Dimple forming at the aft interface

Numerical artifact*:

- Upstream-face dimple

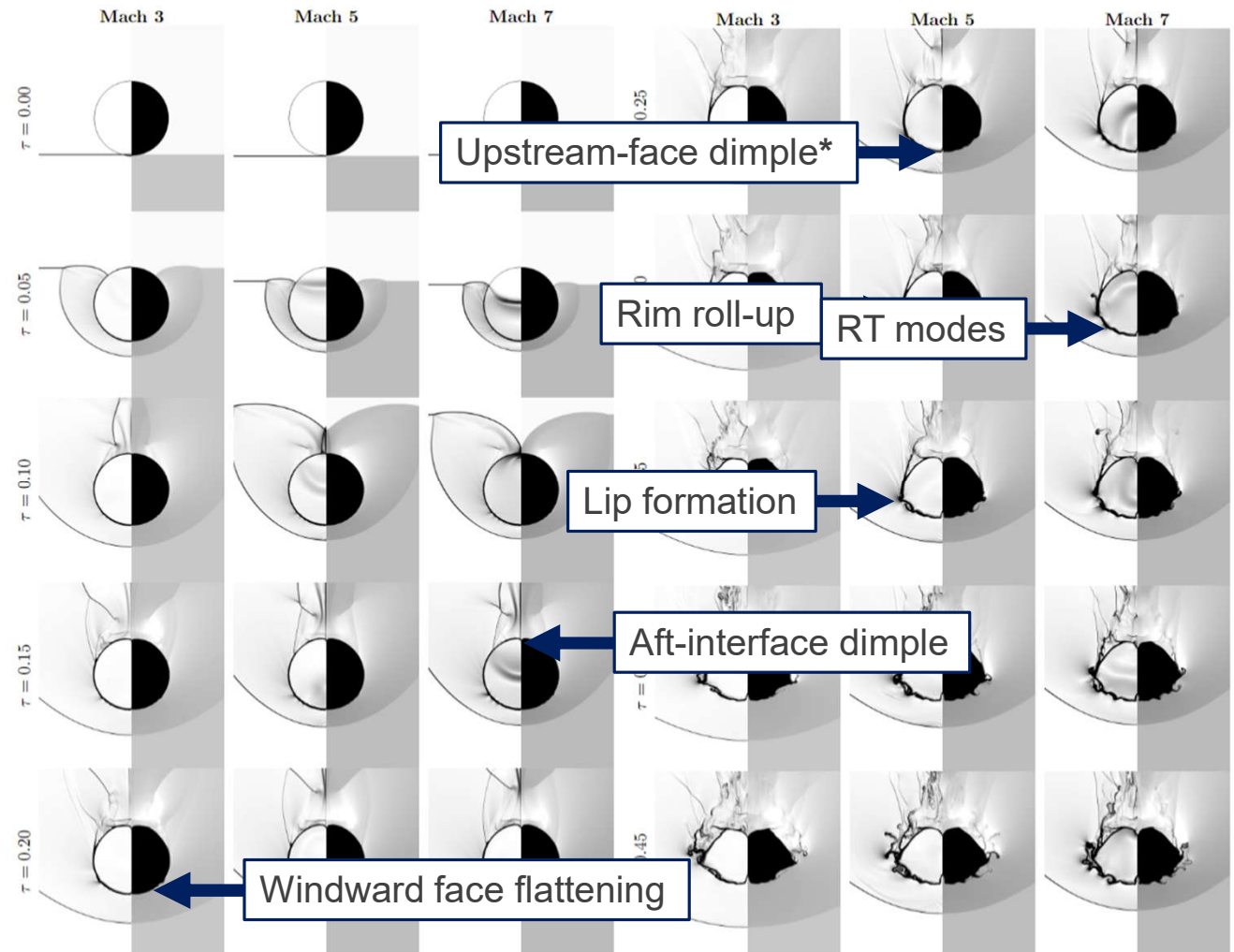


Figure: Numerical schlieren *left* and numerical shadowgraph *right* of the early-time droplet deformation.

Doped Results: Spatial Selectivity of Activation

The normalized underpressure integral predicted that bubble activation should be concentrated in the downstream half of the droplet.

- The simulations confirm this prediction

Consider the Mach 5, $R_0 = 0.2 \mu m$, $N_b = 100$ case:

- Bin the 100 bubbles by initial position

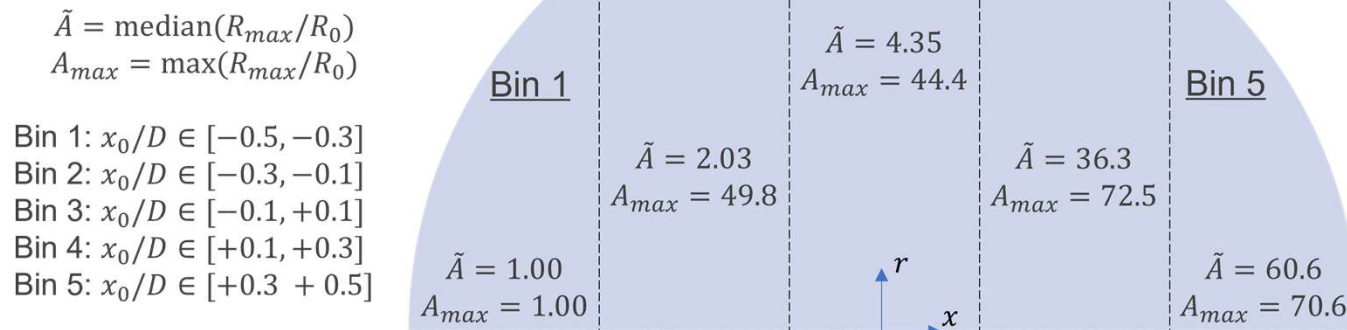
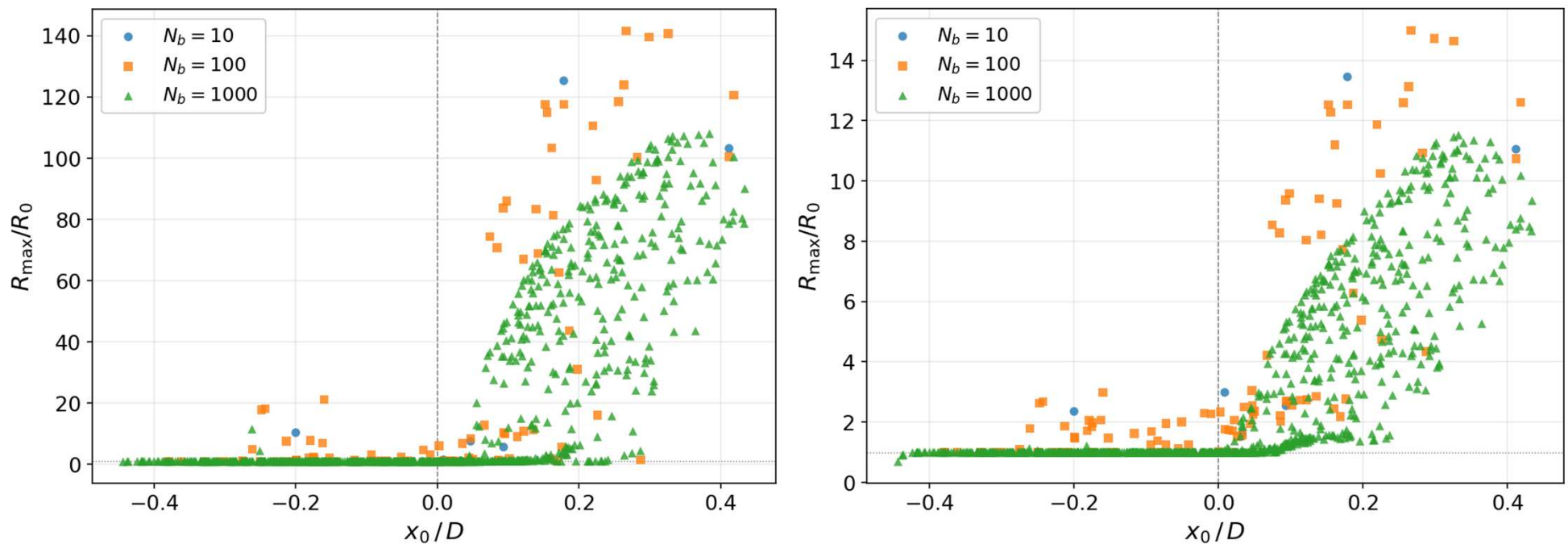


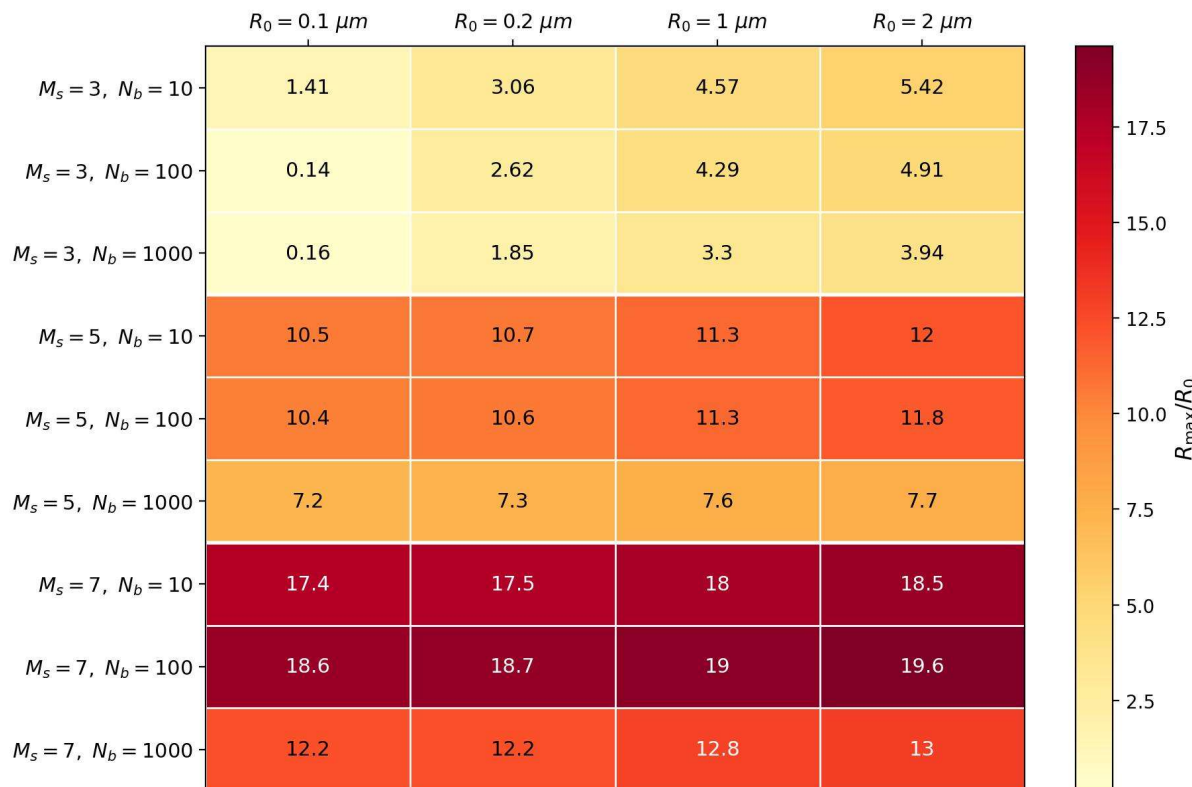
Figure: Spatial bins based on initial streamwise nucleus location, showing median and maximum bubble growth.

Doped Results: Spatial Selectivity of Activation

Downstream half of the droplet shows most bubble activation



Doped Results: Absolute Bubble Growth



Mach number:

- Strongest influence on peak bubble growth ($Mach \uparrow, R_{max} \uparrow$)

Initial bubble radius:

- Significant effect on R_{max}/R_0
- $\sim$ constant R_{abs} for a given Mach

Bubble population:

- Least effect on peak bubble growth
- $N_b = 1000$ seems to limit peak bubble growth when compared to other population sizes

Figure: Absolute maximum bubble radius in μm for all doped cases.

Doped Results: Pressure Perturbations

The volumetric growth and collapse of the Lagrangian bubbles feeds back onto the Eulerian carrier field

$$\Delta P(x, \tau) = P_{doped}(x, \tau) - P_{clean}(x, \tau)$$

The bubble population attenuates the clean acoustic reverberations

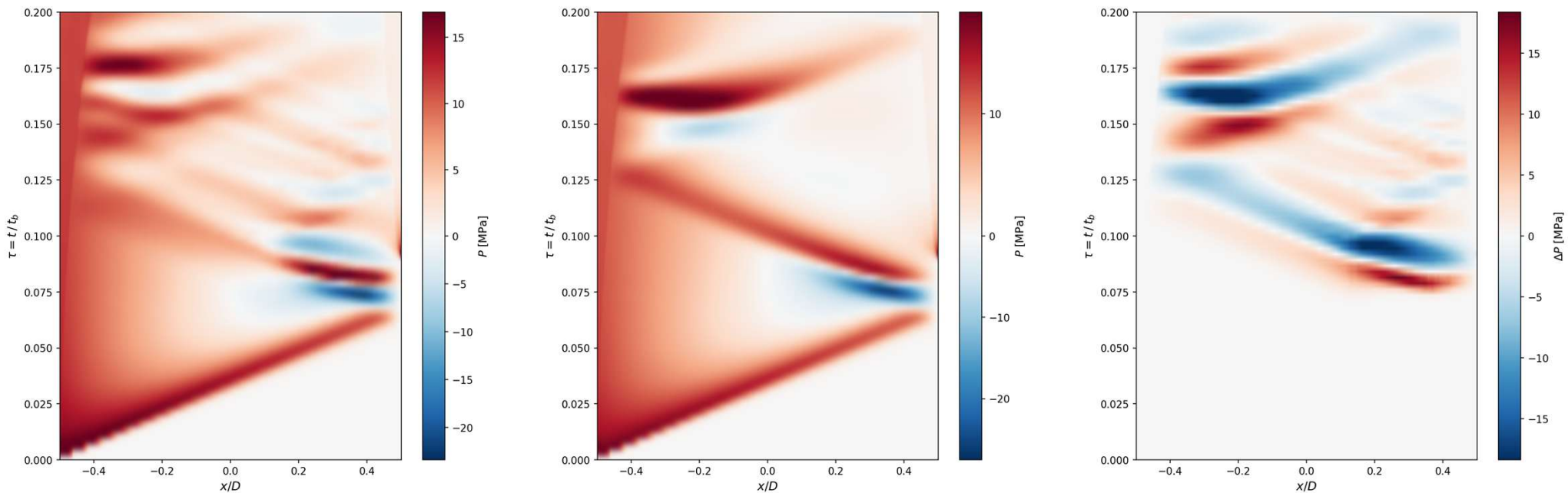


Figure: Space-time diagram of axial centerline pressures at Mach 5. Left: Doped ($R_0 = 0.1 \mu m$, $N_b = 1000$); Center: Clean; Right: pressure difference.

Doped Results: Pressure Perturbations

- The perturbation signal is negligible during the initial shock transmission phase
- The pressure perturbation fills the entire droplet
 - Forms characteristic V-shaped or X-shaped diagonal bands

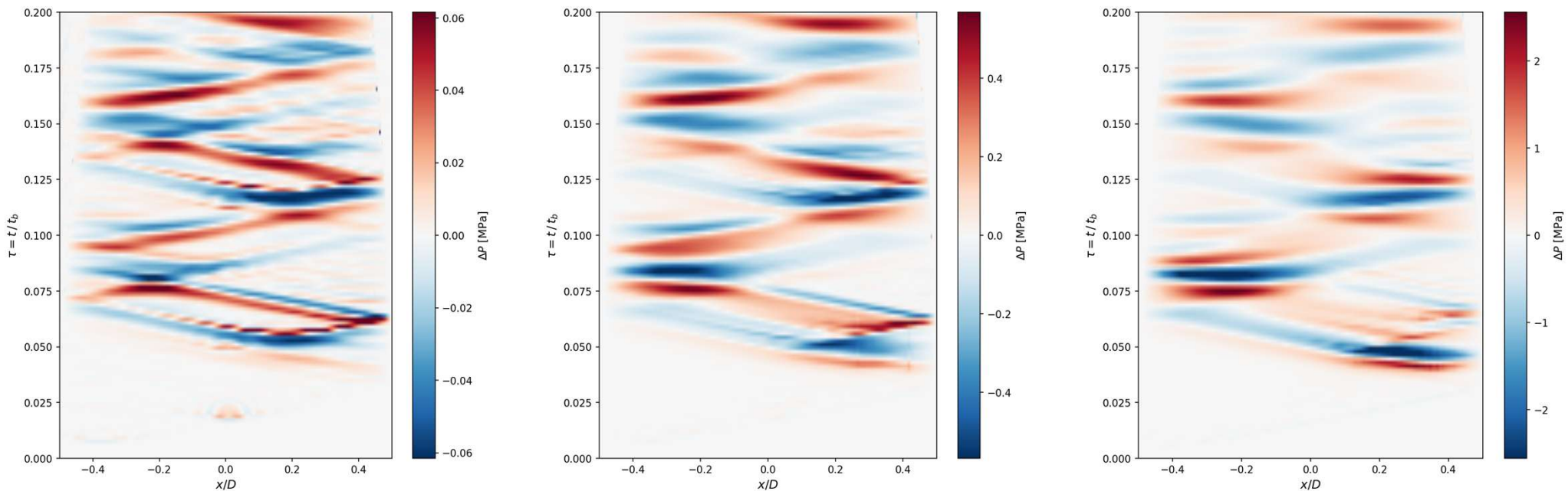
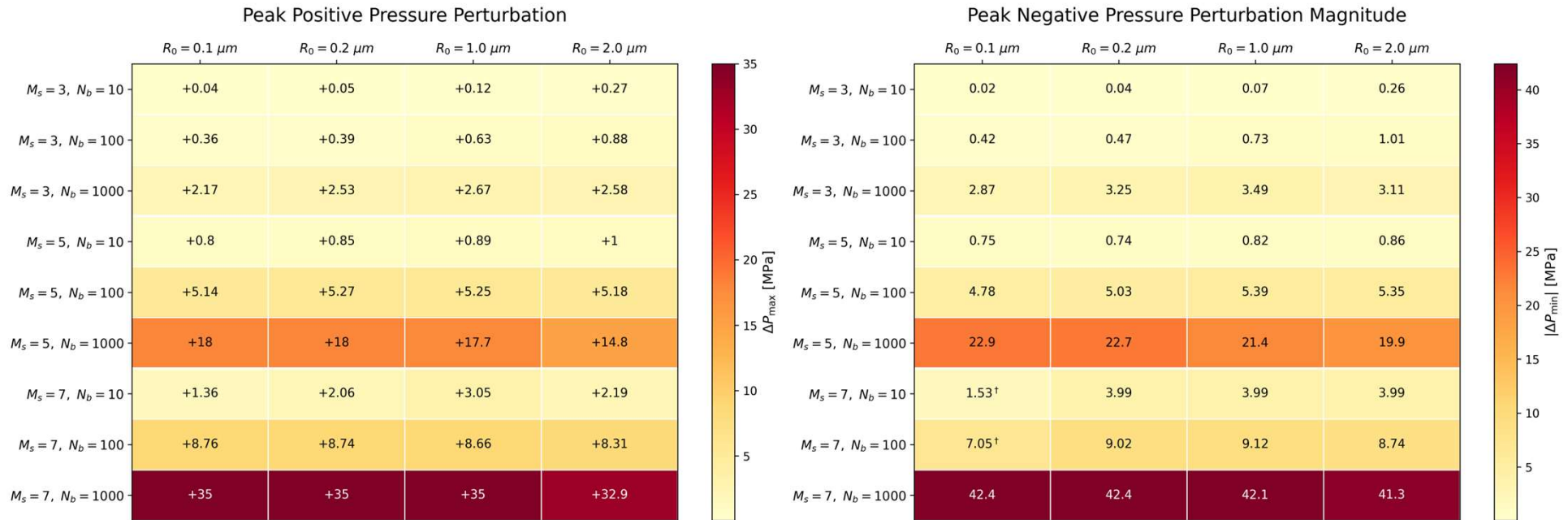


Figure: Space-time diagram of axial centerline pressure differences at Mach 3 at $R_0 = 1.0 \mu\text{m}$.
 Left: $N_b = 10$; center: $N_b = 100$; right: $N_b = 1000$.

Doped Results: Pressure Perturbations



- Peaks are highly dependent on Mach number ($Mach \uparrow, |\Delta P_{max/min}| \uparrow$)
- Peaks are highly dependent on bubble population ($N_b \uparrow, |\Delta P_{max/min}| \uparrow$)

Doped Results: Early Droplet Deformation

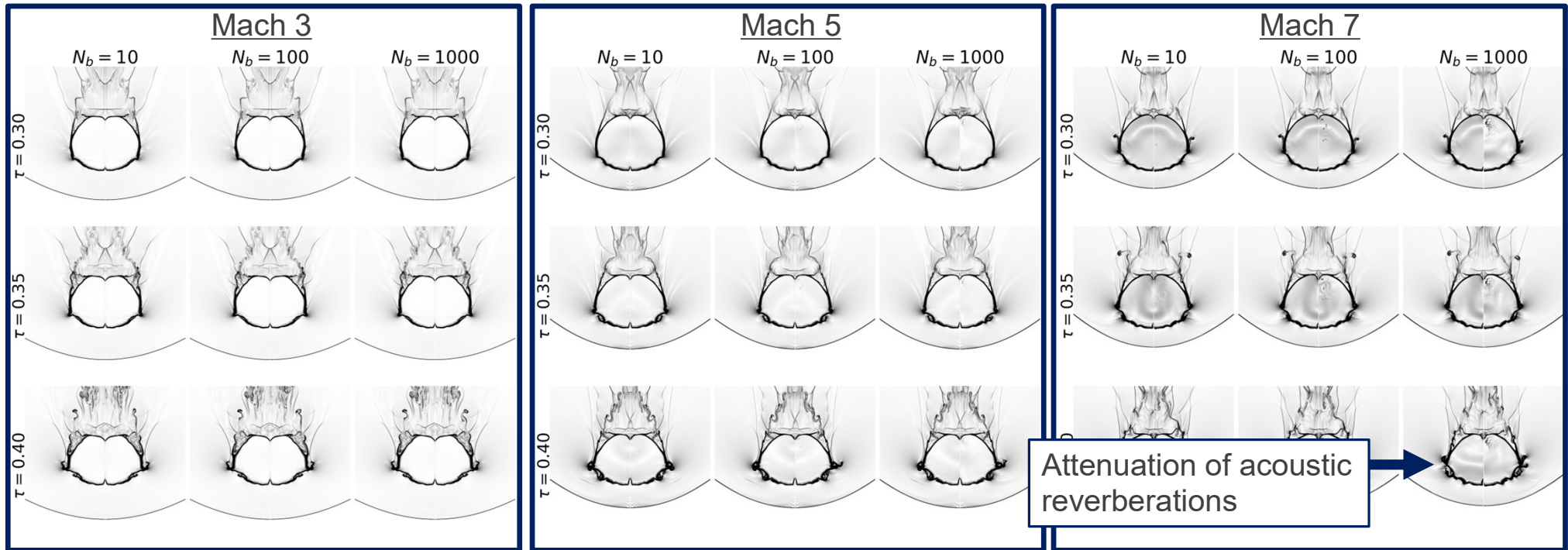


Figure: Numerical schlieren taken at doped cases with $R_0 = 1.0\mu m$. Each frame shows comparison of respective clean half (left) and the doped half (right). *Left: Mach 3; center: Mach 5; right: Mach 7.*

- There is nearly no observable difference* in surface deformation at any Mach or bubble density
 - *Mach 7, $N_b = 1000$, $\tau = 0.40$
- Noticeable damping of internal acoustics in doped cases with $N_b = 1000$

Principal Findings

Main Conclusion:

Pre-existing gas nuclei undergo strong internal activation, but this activation does not measurably alter early droplet-scale deformation over the tested parameter range, given the modeling assumptions.

This result is consistent with Cavainolo 2026 and Briggs 2025 and inconsistent with Shannon 2024.

- Clean-droplet cases establish strong internal acoustic focusing
- Gas nuclei are activated primarily by reflected tensile waves
- Bubble growth is spatially selective
- Dense populations produce acoustic attenuation and self-suppression of individual bubble growth
- Despite strong local bubble dynamics and pressure perturbations, early interface deformation remains nearly unchanged

Contributions and Limitations

Contributions:

- Isolated the purely mechanical response of pre-existing non-condensable gas
- Implemented and verified coupled Eulerian–Lagrangian bubbler dynamics for shock-loaded, bubble-laden droplets
- Benchmarked the Eulerian shock–water framework against shock–water-column experiments
- Mapped bubble activation, pressure feedback, and deformation response across M_s, R_0, N_b

Limitations:

- Vapor production, condensation, and mass transfer were excluded
- Bubble nuclei were fixed in space and did not advect with the liquid
- Axisymmetric assumption limits 3D instability development
- Conclusions are computational and require targeted experimental validation

Conference Paper Submissions

Abstract Submitted: Juan P. Roldan, Brendon A. Cavainolo, Sheryl M. Grace, Michael P. Kinzel, Guillermo A. Jaramillo, “Initial Perturbation Effects on Surface Instability Wavelength Selection in High-Mach Drop Aerobreakup”, AIAA: SciTech, Jan 2027.

Abstract Submitted: Juan P. Roldan, Sutharsan Satcunanathan, Georgios Goinis “POD Airfoil Dimensionality Reduction for Compressor Cascade Optimization”, AIAA: SciTech, Jan 2027.

Manuscript Submitted: Juan P. Roldan, Brendon A. Cavainolo, Rebecca C. Shannon, Sheryl M. Grace, Michael P. Kinzel, “Effect of Entrapped Non-Condensable Gas on Shock-Driven Droplet Deformation: Implications for Hypersonic Rain Environments”, AIAA: International Space Planes and Hypersonic Systems and Technologies Conference, July 2026.

Eric M. Yoerg, Juan P. Roldan, Donald Tuten, Michael Potash, Rui Harrison, Kassandra J. Durst, Michael P. Kinzel, Surabhi Singh “Development of an Exploding Wire Facility to Study Shock Propagation at Embry-Riddle Aeronautical University”, AIAA: SciTech, Jan 2025.

Future Work

Model extensions:

- Add rate-limited phase-change models
- Implement the missing θ -direction stochastic correction
- Allow nuclei to advect with the liquid
- Extend simulations beyond early-time axisymmetric regime
- Extend nuclei modeling to solid/particulate inclusions and heterogeneous cavitation sites

Experimental direction:

- Perform companion experiments with controlled nuclei populations

Broader research path:

- Couple shock-droplet breakup predictions to rain-erosion / surface-damage models
- Extend the framework to propulsion-relevant droplets breakup

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